DMD AND ITS VARIANTS — COMPLETE CLASS MATERIAL

🔴 1. START FROM STANDARD DMD (BASELINE)

We begin with data:

$$x_1, x_2, \ldots, x_m \in \mathbb{R}^n$$

📌 Construct matrices

$$X = [x_1 \; x_2 \; \ldots \; x_{m-1}] \in \mathbb{R}^{n\times(m-1)}$$
$$X' = [x_2 \; x_3 \; \ldots \; x_m] \in \mathbb{R}^{n\times(m-1)}$$

📌 DMD assumption

$$X' = AX$$

📌 Solve for A

$$A = X'X^{\dagger}$$

❗ Problem

If:

- X is large
- noisy
- not full rank

👉 Direct computation becomes unstable

AMRITA VISHWA VIDYAPEETHAM
Dr. Anirban Tarafdar
Assistant Professor
Amrita Viswa Vidyapeetham

🟢 2. EXACT DMD

🔵 Step 1: Apply SVD to X

$$X = U\Sigma V^T$$

Where:

- $U \in \mathbb{R}^{n\times r}$
- $\Sigma \in \mathbb{R}^{r\times r}$
- $V \in \mathbb{R}^{(m-1)\times r}$

🔵 Step 2: Substitute into DMD

$$A = X'(U\Sigma V^T)^{\dagger}$$

Using pseudoinverse property: $X^{\dagger} = V\Sigma^{-1}U^T$

🔵 **Step 3: Final expression**

$$A = X'V\Sigma^{-1}U^T$$

🔵 **Step 4: Reduced operator**

Instead of large A, define:

$$\tilde{A} = U^T AU$$

Substitute:

$$\tilde{A} = U^T X'V\Sigma^{-1}$$

🎯 **Why Exact DMD?**

✓ avoids inversion of large matrix
✓ improves numerical stability
✓ works for high-dimensional data

🎯 **Key takeaway**

👉 "Exact DMD computes dynamics in a reduced subspace."

🟡 **3. TRUNCATED DMD (DIMENSION REDUCTION)**

🔵 **Problem**

Even SVD may keep small noisy components

🔵 **Solution**

Keep only top $r$singular values

$$X \approx U_r\Sigma_r V_r^T$$

🔵 **Compute reduced operator**

$$\tilde{A}_r = U_r^T X' V_r \Sigma_r^{-1}$$

🎯 **Why?**

✓ removes noise
✓ reduces dimension
✓ faster computation

🔢 **Numerical example**

Suppose:

AMRITA VISHWA VIDYAPEETHAM
Dr. Anirban Tarafdar
Assistant Professor
Amrita Viswa Vidyapeetham

$$\Sigma = \begin{bmatrix} 10 & 0 \\ 0 & 0.01 \end{bmatrix}$$

Truncate:

$$\Sigma_r = [10]$$

👉 Ignore small value → noise removed

🎯 **Meaning**

👉 Only dominant dynamics retained

🟣 4. HANKEL DMD (TIME-DELAY EMBEDDING)

🔴 Problem

Standard DMD assumes:

$$x_{k+1} = Ax_k$$

But real system:

$$x_{k+1} = f(x_k, x_{k-1}, \dots)$$

🔵 Solution: Increase dimension

Define:

$$z_k = \begin{bmatrix} x_k \\ x_{k-1} \end{bmatrix}$$

🔵 Construct Hankel matrix

$$H = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ x_2 & x_3 & x_4 & x_5 \end{bmatrix}$$

🔵 Build DMD matrices

$$X = H(:, 1:3), X' = H(:, 2:4)$$

🔵 Compute

$$A = X'X^{\dagger}$$

🔢 Result (from your full calculation)

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$$

AMRITA VISWA VIDYAPEETHAM
Dr. Anirban Tarafdar
Assistant Professor
Amrita Viswa Vidyapeetham

🎯 Why Hankel DMD?

✓ captures memory
✓ converts higher-order → first-order
✓ improves prediction

🔴 5. EXTENDED DMD (EDMD)

🔵 Problem

DMD assumes linear system

🔵 Real system

$$x_{k+1} = x_k^2$$

🔵 Solution: Lift to higher space

Define:

$$\psi(x) = \begin{bmatrix} x \\ x^2 \end{bmatrix}$$

🔵 New system

$$\psi(x_{k+1}) = A\psi(x_k)$$

🔢 Example

Let:

$$x = [1,2,4]$$

Then:

$$\psi(1) = [1,1], \psi(2) = [2,4]$$

Now apply DMD in transformed space

🎯 Why EDMD?

✓ handles nonlinear systems
✓ approximates Koopman operator

AMRITA VIDYAPEETHAM
Dr. Anirban Tarafdar
Assistant Professor
Amrita Viswa Vidyapeetham

🟢 6. SPARSE DMD

🔵 Problem

Too many modes:

$$x_k = \sum b_i \phi_i e^{\lambda_i k}$$

🔵 Solution

Solve:

$$\min_b \; || X - \Phi b ||^2 + \lambda \, || b ||$$

🎯 Meaning

✓ keeps only important modes
✓ removes redundancy

🔢 Example

Modes = 5
Keep only top 2 → better interpretation

⚫ 7. ONLINE / STREAMING DMD

🔵 Problem

Data arrives continuously

🔵 Solution

Update A incrementally:

$$A_{new} = A_{old} + \Delta A$$

🎯 Why?

✓ real-time systems
✓ avoids recomputation

AMRITA VISHWA VIDYAPEETHAM
Dr. Anirban Tarafdar
Assistant Professor
Amrita Viswa Vidyapeetham